Most of the decay will be finished after thirty years, and only about 1 percent of the O_2 will be used up. There is still plenty for Cooper's children and grandchildren to breathe, if they can find anything to eat.

But that 1 percent of the atmospheric O_2 will have been converted into carbon dioxide, which means 0.2 percent of the atmosphere will then be CO_2 (since most of the atmosphere is nitrogen). That's enough CO_2 to make breathing unpleasant for highly sensitive people and perhaps drive the Earth's temperature up (via the greenhouse effect) by 10 degrees Celsius (18 degrees Fahrenheit)—unpleasant for everyone, to put it mildly!

To make everyone's breathing uncomfortable and induce drowsiness, ten times more atmospheric O_2 would have to be converted into CO_2; and to kill most everyone by CO_2 poisoning, an additional five times more would have to be converted, a factor of fifty in all. I have not found a plausible mechanism for this.

So is Professor Brand wrong? (Even theoretical physicists can make mistakes. Especially theoretical physicists. I know; I am one.) Probably yes, he is wrong, but conceivably no. The Professor *could* be right, but it would require geophysicists' understanding of ocean bottoms to be severely flawed.

There is undecayed organic material on the ocean bottoms as well as on land. Geophysicists estimate that the amount on ocean bottoms is about one-twentieth that on land. *If* they are wrong and there is fifty times more on the ocean bottoms than on land, and *if* there is a mechanism to quickly dredge it up, then its decay to produce CO_2 could leave everyone gasping for oxygen and dying of CO_2 poisoning.

Now, once every many thousand years, an instability triggers the ocean to turn over. Water from the surface sinks to the bottom and drives bottom water to the surface. It is conceivable that in Cooper's era there is such an overturn so vigorous that the upwelling bottom water